

Microbiology: Bacterial and Fungal Endocarditis

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Session Objectives

1. Describe the classifications, epidemiology, and risk factors associated with endocarditis.
2. Identify common and atypical bacterial and fungal causes of endocarditis.
3. Describe the pathogenesis associated with different microbial causes of endocarditis.
4. Identify unique microbial features and virulence factors associated with the different pathogens.
5. Outline the laboratory techniques used for the identification and diagnosis of endocarditis.

Definitions

Infective Endocarditis (IE) – **Infection** and inflammation of the heart's **endocardium**, specifically the **valve(s)** of the heart

Classifications

By Onset:

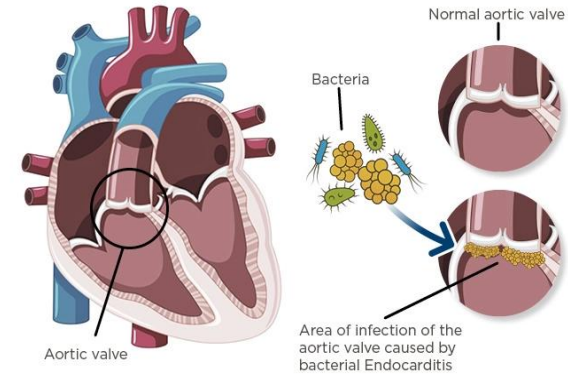
- **Acute IE** – Rapid progression, highly virulent organisms
 - Markedly febrile, lasting days - weeks
- **Subacute IE** – Slow progression, less virulent organisms,
 - Lower fevers, can last for several weeks

By Source:

- Native Valve Endocarditis (**NVE**)
- Prosthetic Valve Endocarditis (**PVE**)
- **IV Drug use-associated IE** – often right sided (Tricuspid valve)

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Endocarditis

↙
Infective

↘
Nonbacterial Thrombotic Endocarditis
(AKA Marantic Endocarditis)

Pathogenesis

Endothelial injury



Formation of
vegetations on
valves consisting of
platelets, fibrin &
microbes



Valve regurgitation
and septic embolism

Risk Factors

- Age >60
- Injection drug use
- Poor dentition/dental infection
- History of infective endocarditis
- Structural heart disease (valvular, congenital)
- Chronic hemodialysis
- Procedures (intravascular device, cardiac implantable electronic device, cardiac bypass surgery)

Epidemiology

- Incidence = 15/100,000 in the US
- More than 50% patients are over the age of 60
- 3x as common in males vs females
- Bacteria are by far the most common cause of endocarditis (fungi rare)

Microbial Causes of IE

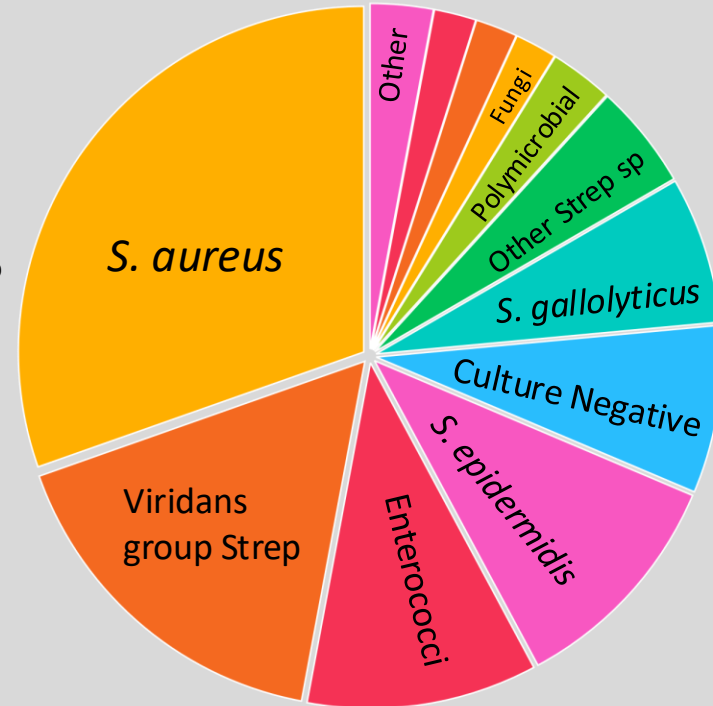
Three **most common causes** of IE worldwide: Staphylococci, Streptococci and enterococci

The distribution of pathogens are as follows:

1. *S. aureus* – 30% of cases
2. Viridans group Streptococci – 16%
3. Enterococci – 11%
4. Coagulase-negative Staphylococci (*S. epidermidis*) – 11%
5. Culture negative endocarditis – 8%
6. *Streptococcus gallolyticus* (Formerly *S. bovis*) – 7%
7. Other streptococci – 5%
8. Polymicrobial – 3%
9. Fungi – 2%
10. Non-HACEK gram-negative bacteria – 2%
11. HACEK Organisms – 2%
12. Other - 3%

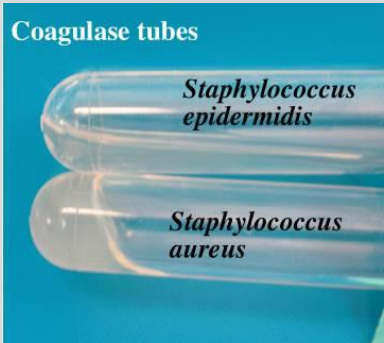
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Staphylococci – aureus and epidermidis

- All Staph spp. are Gram +, **catalase +**, growing in **clumps**
- *S. aureus* is coagulase +
- *S. epidermidis* is novobiocin sensitive



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Feature	<i>S. aureus</i>	<i>S. epidermidis</i>
Gram Stain and Morphology	Gram + Cocci	Gram + Cocci
Coagulase	POSITIVE – clots plasma	NEGATIVE
Biofilm	Rare (not typical)	Strong Biofilm formation , especially on prosthetic devices
Hemolysis	Beta hemolytic (complete)	Gamma Hemolytic (non-hemolytic)
Typical Disease	<u>MCC of Acute</u> Endocarditis (esp. IV Drug users)	<u>Subacute</u> Prosthetic valve endocarditis

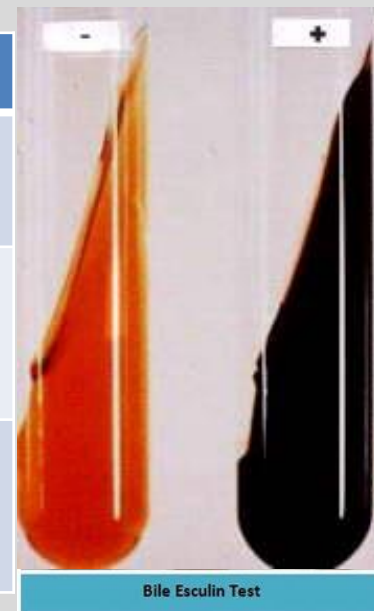
Streptococci – *S. viridans*, *gallolyticus* and *E. faecalis/faecium*

- Catalase test **essential first step** to differentiate Staph vs Strep
- Viridans - Subacute IE from **Dental procedures**
 - Dextran-mediated** adherence to fibrin-platelet aggregates
- Gallolyticus (Bovis) – screen for **colon cancer**

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Species	Lancefield Group	Key Features	Diseases
<i>S. viridans</i> (<i>S. sanguinis</i> , <i>S. mutans</i> , <i>S. mitis</i>)	none	Oral flora	MCC of subacute IE (esp. after dental procedures)
<i>S. gallolyticus</i> (FKA <i>S. Bovis</i>) ~ Bovis in Blood, Cancer in Colon ~	Group D (non-enterococcus)	- Bile esculin + Does not grow in 6.5% NaCl - GI tract	STRONG ASSOCIATION with Colon Ca and GI neoplasms
<i>Enterococcus faecalis/faecium</i>	Group D (enterococcus)	- Bile Esculin + PYR + Grows in 6.5% NaCl - GI/GU tract	Subacute IE after GU/GI procedures



Atypical Gram-Positive Bacteria – NVS, Corynebacteria, Listeria

- Nutritionally Variant Strep (**NVS**) – Oral flora
 - **B6 supplemented media**, satellitism around *S. aureus*
- **Corynebacterium** (diphtheroids) – pleomorphic rods (club shaped) in patients with **prosthetic devices**, Isolated on **Loeffler's medium**
- **Listeria** – elderly/pregnant/immunocompromised patient, **Foodborne** (cold enrichment)

Atypical Gram-Negative Bacteria – HACEK Organisms

- **HACEK** organisms: **MCC of gram-negative endocarditis** in patients that **do not abuse IV drugs**
- All are bacilli found in **normal oral flora**, slow-growing
- Consider **dental disease/oral trauma** as portals of entry

Organism
H aemophilus sp*
A ctinobacillus sp*
C ardiobacterium hominis
E ikenella corrodens
K ingella kingae

* = Haemophilus and Actinobacillus were reclassified into Aggregatibacter Genus

Atypical Gram-negative Bacteria – Coxiella, Legionella, Bartonella

- All are gram negative but **stain poorly** with conventional gram staining
 - **Fastidious or intracellular**, requiring special stains
- Coxiella is the **MCC of culture negative endocarditis**

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Organism	Gram Stain	Morphology	Diagnostic Notes
<i>Coxiella burnetii</i>	Weakly gram negative (often not visible)	Small pleomorphic coccobacilli	• Obligate intracellular • Q fever
<i>Legionella pneumophila</i>	Poorly staining gram negative	Faintly staining bacilli	• Requires Buffered Charcoal Yeast Extract (BCYE) • Legionnaire's disease/Pontiac Fever
Bartonella spp.	Poorly staining gram negative	Pleomorphic bacilli or coccobacilli	• Intracellular in erythrocytes/endothelial cells • Cat scratch (<i>B. henselae</i>), Trench fever (<i>B. quintana</i>)

Virulence Factors Exacerbating Infective Endocarditis

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- 1) **Immune Resistance** – IE causing bacteria can resist complement mediated lysis or phagocytosis
 - **Capsule** protection is a major defense mechanism

- 2) **Adherence to Damaged Valves or Vegetations**
 - **Dextran-Mediated** adherence - extracellular polysaccharide (strep)
 - **FimA protein** - Bacterial surface adhesin (strep)
 - **Fibronectin Binding proteins** - Bacterial surface adhesin (*S. aureus*)

Typical Fungi implicated in IE – Candida, Aspergillus, et al

- Very rare overall, implicated in **invasive heart surgeries**
- **5-15% of prosthetic valve endocarditis & 12-20% of endocarditis in IV drug use disorders**
- Most common causes:
 - *Candida albicans* – a yeast; many people are carriers
 - *Aspergillus* – a ubiquitous mold; not easily culturable
- Less common: *Coccidioides*; *Histoplasma*; & *Cryptococcus*

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↑ Pseudohyphae & budding yeast (Candida)



Fruiting bodies ↑ of ***Aspergillus***
with numerous conidiophores

Bacterial Entry, Spread, and Multiplication

Bacteria/fungi commonly enter bloodstream via:

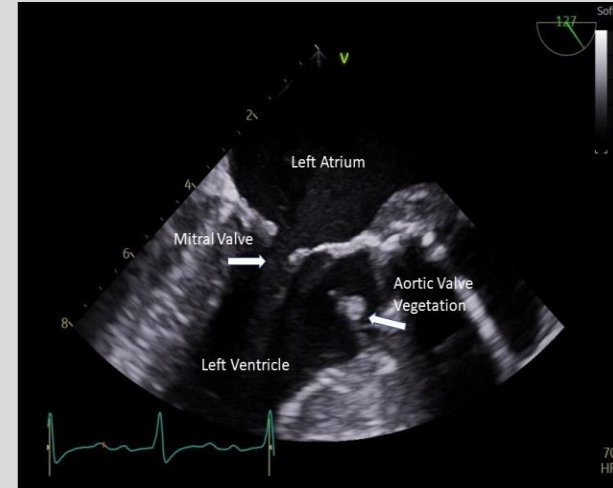
- Trauma to heavily colonized mucosa (dental procedures)
- Indwelling vascular access
- IV Drug use
 - Right sided endocarditis (Tricuspid abnormalities)
 - ***Staph aureus*, *Candida*, *Pseudomonas*** most common

Vegetations:

- Damaged valves become “sticky” from **fibrin and thrombin deposition**, increasing surface area and allows for the deposition of infective organisms, complement and antibodies
- Mitral Valve (Most common overall) > Aortic Valve >> Tricuspid Valve >>>>> Pulmonic valve

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Signs and Symptoms

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♥ Bacteria **FROM JANE** ♥:
Fever
Roth spots
Osler nodes
Murmur
Janeway lesions
Anemia
Nail-bed hemorrhage
Emboli

- Release of **cytokines** =
Systemic Symptoms:
 - Fever, sweats, fatigue,
anorexia, weight loss



- Roth Spot on Retina
- Round white spot
surrounded by
hemorrhage



- Osler node
- **Tender** lesions on
finger/toepads
from immune
complex deposition



- Janeway lesion
- Small, **PAINLESS**
erythematous
lesions on
palm/sole



- Nail bed (splinter)
hemorrhage
- tiny clots that damage
the capillaries under
the nails

Diagnosis

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Modified Duke Criteria

Major Criteria

- Positive blood cultures
- Evidence of endocardial involvement

Minor Criteria

- Predisposing condition or IV drug use
- Fever $> 38^{\circ}\text{C}$
- Vascular phenomena (e.g. emboli, hemorrhage)
- Immunologic phenomena (e.g. Osler's nodes, Roth's spots)
- Blood culture or serologic evidence that does not meet major criterion

Definite infective
endocarditis criteria:

2 major criteria
or
1 major criteria, 3
minor criteria
or
5 minor criteria

- Transthoracic (TTE) or Transesophageal echo (TEE) used to assess for vegetations depending on index of suspicion
- **BLOOD CULTURES BEFORE ABX** whenever possible!

Questions for review- answers and rationale in notes section of PPT

Question 1:

Which of the following scenarios would put someone at the **greatest** risk for developing infective endocarditis?

- A. Antibiotic treatment for an acute sore throat
- B. Bacteremia in a 69-year-old woman with a past history of rheumatic fever or disease
- C. Blood transfusion given during abdominal surgery
- D. Impetigo in a 7-year-old boy
- E. Tooth cleaning of a high school student during a routine dental visit

Question 2:

What two fungal pathogens are most associated with causing endocarditis?

- a. *Candida* and *Aspergillus*
- b. *Candida* and *Histoplasma*
- c. *Candida* and *Cryptococcus*
- d. *Aspergillus* and *Histoplasma*
- e. *Histoplasma* and *Coccidioides*

Question 3:

A 68-year-old woman presents to her family physician with the signs and symptoms of cardiovascular abnormalities. A heart murmur was heard on auscultation, and valvular vegetations were revealed by TTE. Cultures done on blood agar identified the pathogen as being a Gram-Positive cocci growing in clusters that is catalase positive and novobiocin sensitive. Which of the following is the likely cause of this patient's condition?

- A. *Enterococcus faecalis*
- B. *Chlamydophila pneumoniae*
- C. *Staphylococcus epidermidis*
- D. *Bartonella henselae*
- E. *Mycobacterium tuberculosis*

Question 4:

A 66-year-old man is diagnosed clinically with subacute infective endocarditis. Routine blood cultures failed to grow out any organisms until the 10th day of incubation. A gram stain of the culture revealed pleomorphic bacilli that are part of the normal mouth flora and cause <5% of endocarditis cases. The likely pathogen, that is causing this man's disease, is:

- A. *Candida albicans*
- B. One of the HACEK bacteria
- C. Viridans group of *Streptococci*
- D. Coagulase-negative *Staphylococci*
- E. Coagulase-positive *Staphylococci*

Question 5:

Which of the following test procedures is most helpful for making a specific diagnosis of endocarditis with a vegetation?

- A. Bronchoscopy
- B. Chest X-ray
- C. PPD skin test
- D. Transthoracic echocardiography
- E. Urine antigen test

Question 6:

Your 72-year-old male patient appears ill, one month after having open heart surgery where he received a prosthetic heart valve. A diagnostic procedure reveals a small vegetation. A gram positive, catalase negative, bile-esculin positive organism that is able to survive in 6.5% NaCl was identified. The likely etiologic agent is:

- A. Group A *Streptococci*
- B. One of the HACEK bacteria
- C. *Enterococcus faecalis*
- D. *Staphylococcus epidermidis*
- E. *Staphylococcus aureus*

Question 7:

A 37-year-old female with a history of intravenous drug use presents with fever, fatigue, and new onset holosystolic murmur heard loudest at the left lower sternal border that increases with inspiration. A yeast-like organism is isolated from her blood and there is evidence of a vegetation. The most likely affected structure is the:

- A. Tricuspid valve
- B. Pulmonary valve
- C. Pulmonary artery
- D. Mitral Valve
- E. Aortic Valve

Summary

- Various types of microbes (bacterial & fungal) are capable of causing infective endocarditis (IE).
- The major causes include the Staph. and Strep. families, but other less common microbes should be considered.
- The major disease manifestation is a vegetation that typically forms on one of the heart valves.
- If not diagnosed early or if left untreated, IE could lead to serious systemic complications.

References

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